

Employee Attrition Prediction – Executive Summary

This analysis was conducted to understand the key reasons why employees leave the organization and to identify areas where HR can take proactive steps to improve employee retention. The employee data was analyzed to study patterns related to department, job role, salary, work-life balance, years at the company, and other workplace factors.

Key Findings

- The Sales department recorded the highest employee attrition rate (20.63%), indicating that this department requires additional retention efforts.
- Among all job roles, Sales Representatives showed the highest attrition rate (39.76%), making them the highest-risk employee group.
- Employees with lower monthly income were generally more likely to leave, but salary was not the only factor affecting attrition.
- The analysis also showed that overtime, frequent business travel, and specific job roles were strong indicators of employee turnover.
- A Logistic Regression model provided the best overall performance for predicting employee attrition in this project.

Recommendations for HR

- Prioritize retention conversations with employees in the Sales department, particularly Sales Representatives, to understand concerns before they decide to leave.
- Reduce excessive overtime and review frequent business travel schedules to improve work-life balance and employee well-being.
- Provide career development opportunities, recognition programs, and regular employee feedback sessions to improve engagement and job satisfaction.

Conclusion

This project demonstrates that employee attrition is influenced by multiple workplace factors rather than salary alone. The findings can help HR identify employees who may be at a higher risk of leaving and take timely preventive actions. The prediction model should be used as a decision-support tool together with HR expertise, employee feedback, and organizational policies when making retention decisions.